MEEN 305 Final Project

Minimum-Weight Beam Design and Analysis
Tapered Hollow Rectangular Tube in PLA

Team #6

Spring 2026

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1 Problem Statement and Requirements

1.1 Fixed Project Inputs

The beam is simply supported with the following prescribed boundary conditions and loading:

Parameter	Symbol	Value
Support span	L	8.0 in
Total beam length	L_{total}	9.0 in
Load Case 1 magnitude	P_1	5.0 lbf
Load Case 1 location	a_1	4.0 in from support A (midspan)
Load Case 1 orientation	N/A	Upright (load applied vertically)
Load Case 2 magnitude	P_2	8.0 lbf $\quad (2 + \text{Team\#} = 2 + 6)$
Load Case 2 location	a_2	5.0 in from support A $\quad (2 + \frac{6}{2} = 5)$
Load Case 2 orientation	N/A	Rotated 90° about span axis
Required factor of safety	FoS_{min}	≥ 1.5 for both load cases

1.2 Objective

Minimize the weight of the printed beam,

$$\text{minimize} \quad W = \rho \int_0^L A(x)\, dx,$$

subject to the factor-of-safety constraint for both load cases and all engineering and manufacturing constraints defined in Section 5.

1.3 Design Variables

The beam cross-section is a tapered hollow rectangular tube with spanwise lightening openings. The geometry is fully described by **10 continuous design variables**:

Variable	Symbol	Units	Role
Left/mid/right outer width	b_L, b_M, b_R	in	Tube width at $x = 0, L/2, L$
Left/mid/right outer height	h_L, h_M, h_R	in	Tube height at $x = 0, L/2, L$
Wall thickness	t	in	Uniform flange/web thickness
Left/mid/right opening ratio	r_L, r_M, r_R	–	Fraction of inner web height removed

All six profile dimensions (b, h, r) vary continuously along the span via piecewise linear interpolation between the three control points. Wall thickness t is kept spatially uniform, consistent with FDM slice practice.

2 Material Selection

2.1 Available FDM Polymers

The Rapid Prototyping Studio (JCAIN 422) stocks four filament options. Their nominal bulk properties and effective FDM properties are:

Material	E [psi]	σ_{allow} [psi]	ρ [lbf/in^3]	k_{orient}	k_{infill}	E_{eff} [psi]
PLA	500 000	4 000	0.045	0.70	0.85	297 500
ASA	380 000	3 600	0.043	0.65	0.85	209 950
PETG	290 000	3 500	0.046	0.72	0.85	177 480
TPU	10 000	2 500	0.043	0.75	0.90	6 750

2.2 Orientation and Infill Factors

FDM parts are mechanically anisotropic because deposited layers bond through partial re-melting. The interlayer bond is the dominant failure surface. Two reduction factors are applied multiplicatively to all nominal isotropic properties:

- **Orientation factor** k_{orient}: accounts for the reduction in strength and stiffness perpendicular to the print plane. When the span axis lies along the print direction (layer lines parallel to the neutral axis), load is transferred through the solid infill struts rather than across layer interfaces. We use $k_{orient} = 0.70$ for PLA, consistent with RPS guidance that layer lines are "the single biggest failure point."
- **Infill factor** k_{infill}: accounts for the reduced solid volume at less-than-100% rectilinear infill. We use $k_{infill} = 0.85$, corresponding to approximately 40% rectilinear infill with solid top/bottom shells. The physical beams were actually printed at 100% infill density (confirmed from the slicer project file), making the fabricated parts stiffer and stronger than the model predicts. This means measured deflections may be somewhat lower than the theoretical values, and the actual factor of safety exceeds the computed minimum.

Effective properties:

$$E_{eff} = k_{orient}\, k_{infill}\, E_{bulk} = 0.70 \times 0.85 \times 500\,000 = 297\,500 \text{ psi}$$

$$\sigma_{allow,eff} = k_{orient}\, k_{infill}\, \sigma_{allow} = 0.70 \times 0.85 \times 4\,000 = 2\,380 \text{ psi}$$

2.3 Justification for PLA

PLA is selected for the following engineering reasons:

1. **Highest effective stiffness**: $E_{eff,PLA} = 297\,500$ psi versus 209 950, 177 480, and 6 750 psi for ASA, PETG, and TPU respectively. Deflection is inversely proportional to $E_{eff}I$; PLA permits the smallest cross-section for a given deflection.
2. **Highest allowable stress**: $\sigma_{allow,eff,PLA} = 2\,380$ psi gives the largest FoS margin for a given cross-section, enabling more material removal.
3. **Printability**: PLA is the recommended default at the RPS and requires no special enclosure or bed adhesion, making reliable prints more likely.
4. **Density**: PLA density (0.045 lbf/in^3) is representative of structural polymers; ASA is marginally lighter but at the cost of significantly lower effective properties.

TPU is immediately eliminated (flexible elastomer, not a structural polymer). ASA and PETG are eliminated because both have lower E_{eff} and lower $\sigma_{allow,eff}$ than PLA; the same geometry would be heavier and weaker.

3 Beam Parametrization

3.1 Cross-Section Topology

At each spanwise location x, the cross-section is a hollow rectangular tube with a centered web opening (lightening slot) in both vertical web faces. The outer dimensions at position x are $b(x) \times h(x)$; the wall thickness t is uniform throughout. The opening removes a fraction $r(x)$ of the clear inner web height $h_{clear}(x) = h(x) - 2t$ from both web faces, leaving solid flanges top and bottom.

3.2 Profile Functions

Each of the three continuous profiles is defined by piecewise linear interpolation between three control points at $x = 0$ (left support), $x = L/2 = 4$ in (midspan), and $x = L = 8$ in (right support):

$$b(x) = \begin{cases} b_L + \dfrac{2(b_M - b_L)}{L}\, x, & 0 \le x \le \frac{L}{2} \\ b_M + \dfrac{2(b_R - b_M)}{L}\,(x - \frac{L}{2}), & \frac{L}{2} < x \le L \end{cases}$$

and analogously for $h(x)$ and $r(x)$. This three-point piecewise-linear scheme provides enough shape freedom to represent arch profiles, linearly tapered beams, and uniform beams as special cases, while keeping the number of design variables tractable.

3.3 Section Properties

Cross-sectional area (with openings).

$$A(x) = 2\,b(x)\,t + [1 - r(x)]\,2\,t\,h_{clear}(x), \quad h_{clear}(x) = h(x) - 2t$$

The first term is the two horizontal flanges; the second is the remaining vertical web area after the lightening slot is removed.

Second moment of area: upright bending (Load Case 1). The full hollow-tube second moment minus the two opening slots:

$$I_{upright}(x) = \frac{b(x)\,h(x)^3 - [b(x) - 2t]\,[h(x) - 2t]^3}{12} - 2\,\frac{t\,[r(x)\,(h(x) - 2t)]^3}{12}$$

Second moment of area: side-rotated bending (Load Case 2). The beam is physically rotated 90° about the span axis, so b becomes the vertical (bending) dimension and h becomes the lateral dimension:

$$I_{side}(x) = \frac{h(x)\,b(x)^3 - [h(x) - 2t]\,[b(x) - 2t]^3}{12} - 2\,\frac{t\,[r(x)\,(b(x) - 2t)]^3}{12}$$

Section modulus. The section modulus uses $c_y(x)$ for upright bending (extreme fiber in the y-direction, i.e. half the outer height) and $c_z(x)$ for side-rotated bending (extreme fiber in the z-direction, i.e. half the outer width):

$$S_{upright}(x) = \frac{I_{upright}(x)}{c_y(x)}, \qquad c_y(x) = \frac{h(x)}{2}$$

$$S_{side}(x) = \frac{I_{side}(x)}{c_z(x)}, \qquad c_z(x) = \frac{b(x)}{2}$$

Shear area.

$$A_{shear}(x) = [1 - r(x)]\, 2t\, h_{clear,active}(x)$$

where $h_{clear,active}$ is the clear inner dimension in the active bending orientation: $h_{clear}(x) = h(x) - 2t$ for upright, $(b(x) - 2t)$ for side.

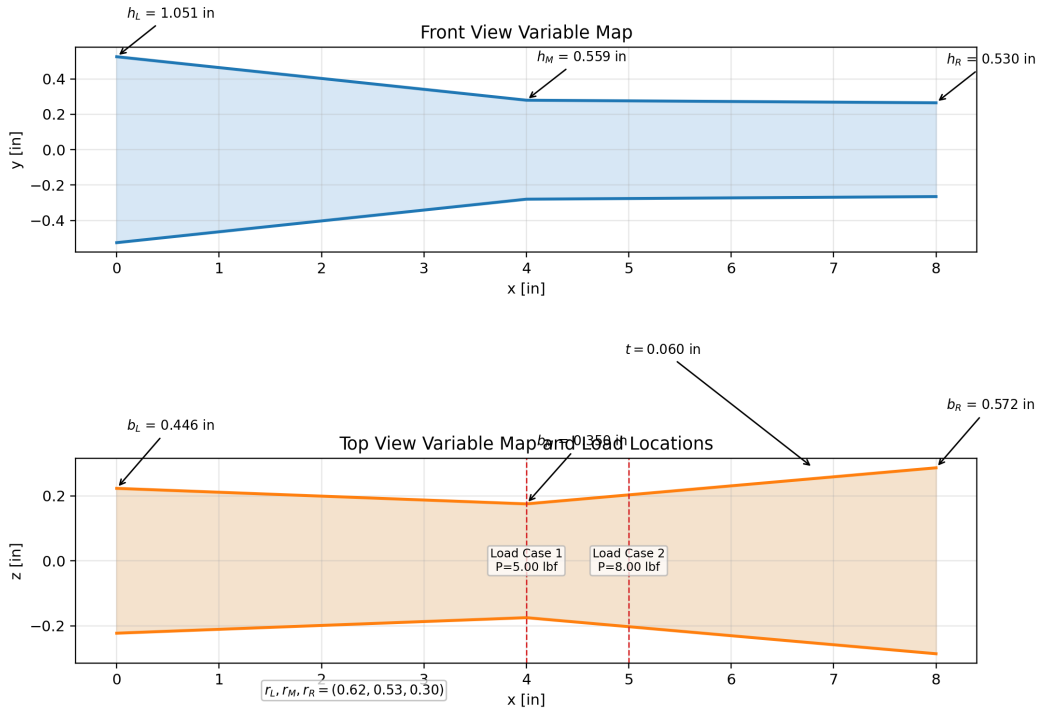

Figure 1: Variable map for the final optimized design. Front view (top) shows the height profile $h(x)$ with control-point annotations; top view (bottom) shows the width profile $b(x)$, wall thickness t, and load locations.

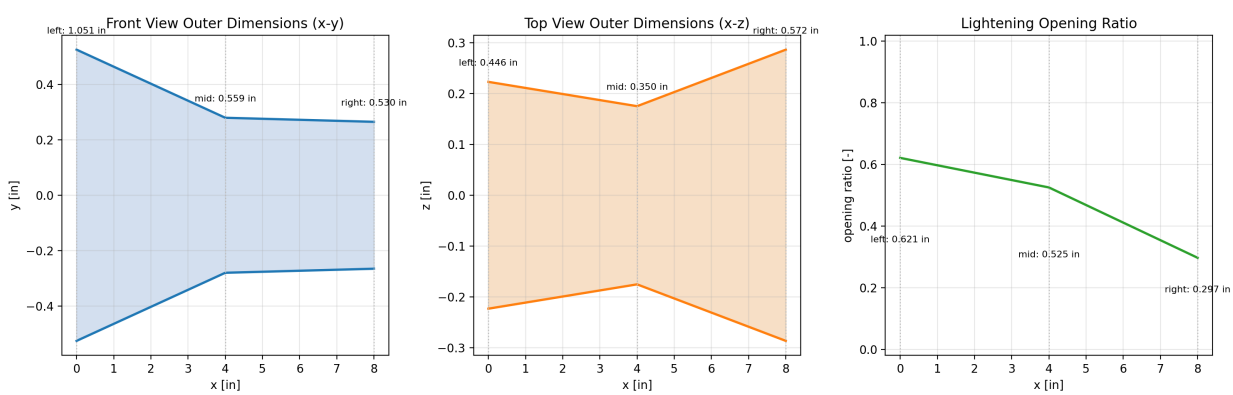

Figure 2: Outer dimension profiles and lightening-opening ratio for the final design. Left: front-view height profile $h(x)$; centre: top-view width profile $b(x)$; right: web opening ratio $r(x)$ along the span.

4 Governing Equations and Analysis Methodology

4.1 Statics: Reactions and Internal Loads

For a simply supported beam with one point load P at $x = a$:

$$R_A = \frac{P(L-a)}{L}, \qquad R_B = \frac{Pa}{L}$$

Shear force and bending moment (piecewise, with the load-point discontinuity):

$$V(x) = \begin{cases} R_A & 0 \le x < a \\ R_A - P & a < x \le L \end{cases} \qquad M(x) = \begin{cases} R_A\, x & 0 \le x < a \\ R_A\, x - P(x-a) & a < x \le L \end{cases}$$

4.2 von Mises Stress

The von Mises criterion combines all stress components into one scalar that can be compared directly against the effective allowable stress. The three intermediate quantities that feed into it are:

Normal bending stress. Using the Euler-Bernoulli flexure formula at the extreme fiber (distance c_y or c_z from the neutral axis, depending on orientation):

$$\sigma_b(x) = \frac{|M(x)|}{S(x)}$$

Shear stress. Shear is distributed over the active web area remaining after the lightening opening. A conservative uniform-distribution approximation is used:

$$\tau(x) = \frac{|V(x)|}{A_{shear}(x)}$$

von Mises equivalent stress. For the plane-stress state at the critical point (bending-dominant at the extreme fiber, shear-dominant at the neutral axis):

$$\sigma_{vm}(x) = \sqrt{\sigma_b(x)^2 + 3\,\tau(x)^2}$$

The σ_b^2 term dominates wherever $|M(x)|$ is large (e.g. midspan); the $3\tau^2$ term dominates near the supports where $M \to 0$ but V is nonzero. The governing (maximum) von Mises stress determines the minimum FoS over the span.

4.3 Factor of Safety

$$\mathrm{FoS}(x) = \frac{\sigma_{allow,eff}}{\sigma_{vm}(x)}$$

The governing (minimum) FoS is tracked at every discretized station x across both load cases:

$$\mathrm{FoS}_{min} = \min_{x,\,\mathrm{LC}} \mathrm{FoS}(x)$$

The design constraint is $\mathrm{FoS}_{min} \ge 1.5$.

4.4 Deflection via Euler-Bernoulli Integration

The Euler-Bernoulli beam equation for a non-uniform cross-section:

$$E_{eff}\, I(x)\, v''(x) = M(x)$$

Because $I(x)$ varies continuously with x, slope $\theta(x) = v'(x)$ and deflection $v(x)$ are obtained by numerical integration (trapezoidal rule, 801 stations) with simple-support boundary conditions $v(0) = v(L) = 0$:

$$\theta(x) = \int_0^x \frac{M(s)}{E_{eff}\, I(s)}\, ds + C_1$$
$$v(x) = \int_0^x \theta(s)\, ds + C_1\, x$$

The integration constant C_1 is determined by enforcing $v(L) = 0$.

4.5 Beam Weight

$$W = \rho \int_0^L A(x)\, dx$$

The integral is evaluated numerically over the same 801-station grid.

4.6 Assumption: Self-Weight Neglected

Self-weight is not included in the shear, moment, stress, or deflection calculations. Justification:

- The optimized beam weighs $W = 0.03454$ lbf.
- Applied point loads total $P_1 + P_2 = 5 + 8 = 13$ lbf.
- Self-weight is $0.03454/13 \approx 0.27\%$ of applied load.
- Standard practice neglects distributed self-weight when it is less than 5% of applied load. At 0.27%, the assumption is conservative by more than an order of magnitude; including self-weight would increase midspan stress by $< 0.3\%$.

4.7 Equation Dependency Chain

The full analysis pipeline from inputs to objective is:

$$(L,\, P,\, a) \xrightarrow{\text{statics}} R_A,\, R_B \;\rightarrow\; V(x),\, M(x) \xrightarrow{\text{geometry}} A(x),\, I(x),\, S(x) \;\rightarrow\; \sigma_b,\, \tau \;\rightarrow\; \sigma_{vm} \;\rightarrow\; \text{FoS}$$

$$M(x),\, E_{eff},\, I(x) \xrightarrow{\text{integration}} v(x) \qquad\qquad A(x),\, \rho \xrightarrow{\text{integration}} W$$

Both load cases use the same geometry functions; the active bending orientation switches between $I_{upright}$, c_y (LC1) and I_{side}, c_z (LC2).

5 Engineering Judgment Constraints

An idealized stress analysis would permit geometries that are physically unrealizable on an FDM printer. The following engineering judgment constraints are hard-coded in `beam_solver.py` (`TaperedRectangularT`

1. **Minimum cross-section at every station, including supports.** At a simply supported beam end, the bending moment is zero, so idealized theory requires zero section modulus (i.e., zero cross-section). In reality, the beam must physically contact its supports and transfer shear. The code enforces $h(x), b(x) \geq 0.15$ in (≈ 3.8 mm) at every station.

2. **Minimum printable wall thickness.** A purely stress-based analysis would permit paper-thin walls wherever stress is low. On a 0.4 mm nozzle, fewer than two continuous perimeters (< 0.8 mm) produces unreliable, porous walls. The code enforces $t \geq 0.040$ in (≈ 1.0 mm).

3. **Wall thickness must fit inside the cross-section.** $t < b/2$ and $t < h/2$ at every station; otherwise the tube wall would overlap the centerline.

4. **Ligament constraint for web openings.** The remaining solid ligament on each side of a web opening must exceed the wall thickness: $\frac{1}{2}(1-r)\, h_{clear} > t$. This ensures the web-to-flange junction is thick enough for load transfer and printing.

5. **Opening ratio bounded below 0.95.** A ratio of 1.0 would remove the web entirely, leaving two unconnected flanges. The code enforces $r < 0.95$.

The **optimizer search bounds** are more restrictive than these hard minimums, providing the primary engineering envelope:

Variable	Optimizer minimum	Optimizer maximum
Height h [in]	0.38	1.30
Width b [in]	0.35	1.60
Wall thickness t [in]	0.06	0.18
Opening ratio r [–]	0.00	0.87

The minimum height of 0.38 in (≈ 9.7 mm) ensures a realistic cross-section at every station including at the supports. The maximum opening ratio of 0.87 was chosen to guarantee at least 1.9 mm of solid ligament per side at the minimum-height station, about 5 nozzle widths at 0.4 mm, sufficient for a reliable continuous wall.

6 Optimization Methodology

6.1 Search Strategy

A two-phase surrogate-free method was used:

1. **Latin hypercube sampling (LHS).** 600 candidate designs are drawn by stratified random sampling across the 10-dimensional design space. LHS guarantees that each variable's range is sampled uniformly, avoiding clustering inherent in purely random Monte Carlo sampling.

2. **Elite local refinement.** After the global phase, the top 20 feasible designs ("elites") are identified. Around each elite, 35 new candidates are sampled from a Gaussian neighborhood with standard deviation shrinking by $1/(k+1)$ each round. This is repeated for 8 rounds, progressively concentrating the search near the current best design.

Total designs evaluated: **1 300**; feasible designs found: **684**.

6.2 Objective Function

The search minimizes a penalized composite objective:

$$f(\mathbf{d}) = W + 500\left(\max(0,\ 1.5 - \mathrm{FoS}_{min})\right)^2 + 12.0\left(\mathrm{FoS}_{min} - 1.5\right)^2 + 0.5\,\overline{u}^2$$

where:

- W is beam weight [lbf], which is the true objective.
- The first penalty drives infeasible designs toward FoS ≥ 1.5.
- The third term (weight 12.0) penalizes *excess* FoS above 1.5; this is what drives the optimizer to remove material rather than stopping at the first feasible design.
- $\overline{u}^2$ is the mean-squared underutilization across beam stations, penalizing cross-sections far from the target stress level and encouraging a more uniform stress distribution.

The excess-FoS weight of 12.0 was determined through systematic trial. Values below ~ 1.0 left the optimizer content with FoS $\gg 1.5$; values above ~ 20 caused constraint violations.

6.3 Why Load Case 2 Governs for Team 6

For Team 6, $P_2 = 8$ lbf at $a_2 = 5$ in produces a maximum bending moment of:

$$M_{max,LC2} = R_{A,LC2} \times a_2 = \frac{8(8-5)}{8} \times 5 = 3 \times 5 = 15 \text{ lbf} \cdot \text{in}$$

Load Case 1 produces:

$$M_{max,LC1} = R_{A,LC1} \times a_1 = \frac{5(8-4)}{8} \times 4 = 2.5 \times 4 = 10 \text{ lbf} \cdot \text{in}$$

LC2 has 50% more bending moment than LC1, and acts in the side-rotated orientation where the width b (not height h) is the critical bending dimension. This fundamentally changes the optimal design topology compared to a team with a smaller P_2: the optimizer must now ensure adequate *width* at the location of peak LC2 moment.

6.4 Printability Bounds

Early runs produced designs with extreme height tapers ($h_{max}/h_{min} > 4\times$), which generated non-contiguous 3D meshes. Capping the maximum height at 1.30 in limits the taper ratio to $\leq 2.13\times$ in the final design, ensuring a single contiguous 3D solid.

7 Final Optimized Design

7.1 Geometry

Variable	Value	Notes
b_L [in]	0.823	Wide left end; carries LC2 side bending and LC1 shear
b_M [in]	0.350	Minimum allowed; width is not the critical dimension at midspan
b_R [in]	0.350	Minimum allowed; LC2 moment decreases toward right support
h_L [in]	0.959	Tall left end provides LC1 upright bending capacity and web shear area
h_M [in]	0.727	Moderate midspan height; LC1 upright bending governs here
h_R [in]	0.606	Shorter right end; LC2 reaction is large at this support ($R_B = 5$ lbf)
t [in]	0.060	Minimum allowed (≈ 1.5 mm, 3.75 nozzle widths)
r_L [–]	0.334	33% web opening at left (limited by LC2 shear demand)
r_M [–]	0.650	65% web opening at midspan where both moments are moderate
r_R [–]	0.276	28% web opening at right where R_B is largest

Topology. The optimizer found a **left-heavy taper**: the left end is the widest ($b_L = 0.823$ in) and tallest ($h_L = 0.959$ in), while the beam tapers down toward the right. This makes physical sense:

- LC2 moment peaks at $x = 5$ in (load location) and increases monotonically from $x = 0$ on the left side. For side-rotated bending, section modulus $S_{side} \propto b^2$; the optimizer responds by making b largest on the left where moment is increasing.
- LC1 midspan bending ($x = 4$ in, upright) is entirely handled by the height profile; $h_M = 0.727$ in gives $\mathrm{FoS}_{LC1} = 3.96$, safely above 1.5 but not the governing case.
- The large right reaction $R_{B,LC2} = 5$ lbf requires solid material at the right end to transfer shear, so $r_R = 0.276$ (only 28% opening) versus $r_M = 0.650$ (65% opening at midspan where both moments are moderate).

Printability check. The minimum remaining ligament per web side is at the midspan station:

$$\ell_{min} = \tfrac{1}{2}(1 - r_M)(h_M - 2t) = \tfrac{1}{2}(1 - 0.650)(0.727 - 0.120) = 0.1062 \text{ in} \approx 2.70 \text{ mm}$$

This is ≈ 6.7 nozzle widths at 0.4 mm, providing a continuous reliable wall. Height taper ratio:

$$h_{max}/h_{min} = 0.959/0.606 = 1.58\times$$

Well below the 4.73× ratio of early non-printable designs.

7.2 Profile Equations

$$b(x) = \begin{cases} 0.8230 + (-0.2365)\,x, & 0 \le x \le 4.0000 \\ 0.3500 + (0.0000)\,(x - 4.0000), & 4.0000 < x \le 8.0000 \end{cases}$$

$$h(x) = \begin{cases} 0.9594 + (-0.0582)\,x, & 0 \le x \le 4.0000 \\ 0.7266 + (-0.0302)\,(x - 4.0000), & 4.0000 < x \le 8.0000 \end{cases}$$

$$r(x) = \begin{cases} 0.3342 + (0.0791)\,x, & 0 \le x \le 4.0000 \\ 0.6504 + (-0.1872)\,(x - 4.0000), & 4.0000 < x \le 8.0000 \end{cases}$$

7.3 Section Properties Profiles

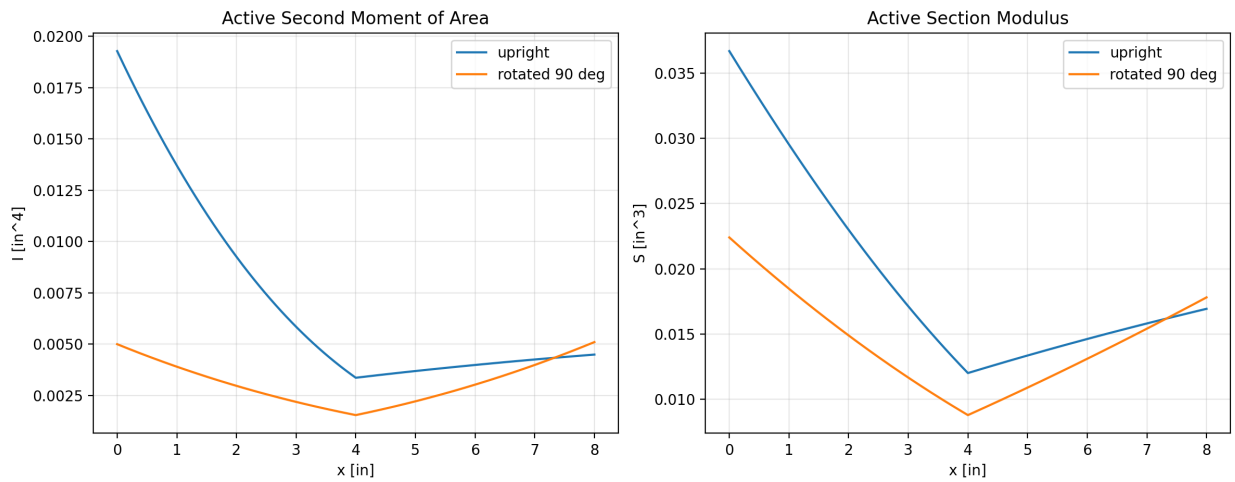

Figure 3: Second moment of area $I(x)$ (left) and section modulus $S(x)$ (right) for the final design in both upright and side-rotated orientations. Note that I_{side} and S_{side} are largest near the left support where $b_L = 0.823$ in; the optimizer places width where LC2 bending moment is largest (increasing from left to the load at $x = 5$ in).

8 Load Case Analysis Results

8.1 Summary Table

Table 1: Load case analysis summary for the final optimized design. LC2 governs with a minimum FoS of 1.522 at $x = 5.0$ in.

Load Case	Orientation	P [lbf]	a [in]	R_A [lbf]	R_B [lbf]	Min FoS	Max $\lvert v \rvert$ [in]
LC1	Upright	5.000	4.000	2.500	2.500	3.956	0.02538
LC2	Side-rotated	8.000	5.000	3.000	5.000	**1.522**	0.11608

8.2 Governing Load Case

Load Case 2 governs with $\text{FoS}_{min} = 1.522$ at $x = 5.0$ in. This is the load application point for LC2, exactly where $M_{LC2}(x)$ reaches its maximum of 15 lbf · in, which is 50% larger than the LC1 peak moment of 10 lbf · in. For Team 6, LC2 is the structurally demanding case because the load is heavier (8 vs 5 lbf), applies at an off-center location that generates a larger moment arm, and acts in the side-rotated orientation where the critical section modulus $S_{side} \propto b^2$ requires large width.

8.3 Load Case 1: Upright Bending

Reactions:

$$R_A = \frac{5.000(8.000 - 4.000)}{8.000} = 2.500 \text{ lbf}, \qquad R_B = \frac{5.000 \times 4.000}{8.000} = 2.500 \text{ lbf}$$

$$\text{FoS}_{min,LC1} = \frac{\sigma_{allow,eff}}{\sigma_{vm,max,LC1}} = \frac{2\,380}{602} = 3.956 \quad \text{at } x = 4.000 \text{ in}$$

Load Case 1 is easily satisfied because the design must be sized for the larger LC2 load; LC1 becomes a secondary condition with comfortable margin.

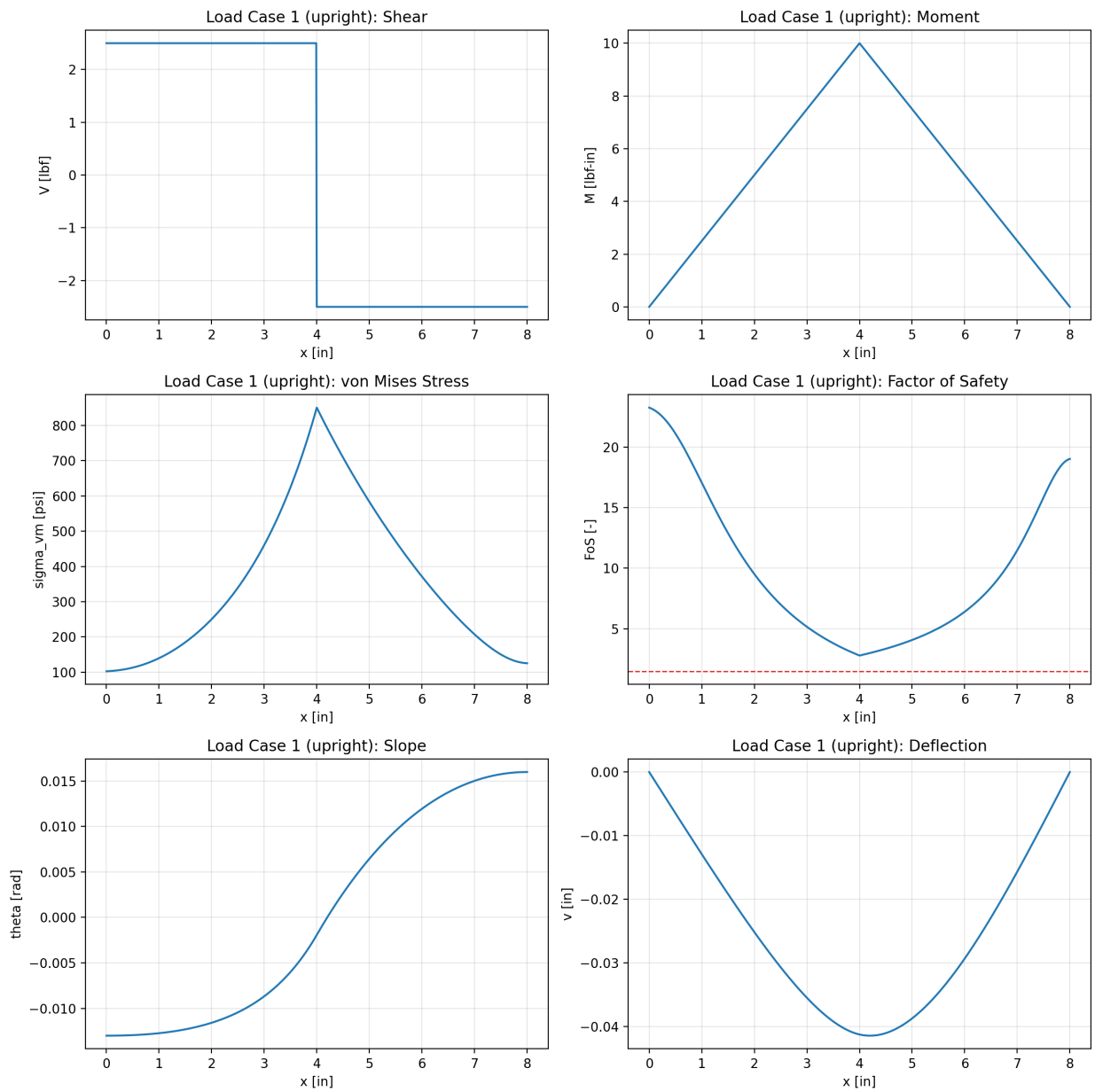

Figure 4: Load Case 1 results (upright orientation, $P_1 = 5$ lbf, $a_1 = 4$ in). Clockwise from top-left: shear diagram, bending moment diagram, von Mises stress, factor of safety (dashed red line at FoS = 1.5), slope, and deflection. The governing FoS of 3.956 at midspan showing significant reserve capacity under LC1; the design is sized by the heavier LC2.

8.4 Load Case 2: Side-Rotated Bending (Governing)

Reactions (Team 6: $P_2 = 8$ lbf, $a_2 = 5$ in):

$$R_A = \frac{8.000(8.000 - 5.000)}{8.000} = 3.000 \text{ lbf}, \qquad R_B = \frac{8.000 \times 5.000}{8.000} = 5.000 \text{ lbf}$$

$$\text{FoS}_{min,LC2} = \frac{\sigma_{allow,eff}}{\sigma_{vm,max,LC2}} = \frac{2\,380}{1\,564} = 1.522 \quad \text{at } x = 5.000 \text{ in}$$

The beam is physically rotated 90° about the span axis, so bending now occurs about the width axis. $I_{side}(x)$ and $S_{side}(x)$ are used in place of the upright section properties.

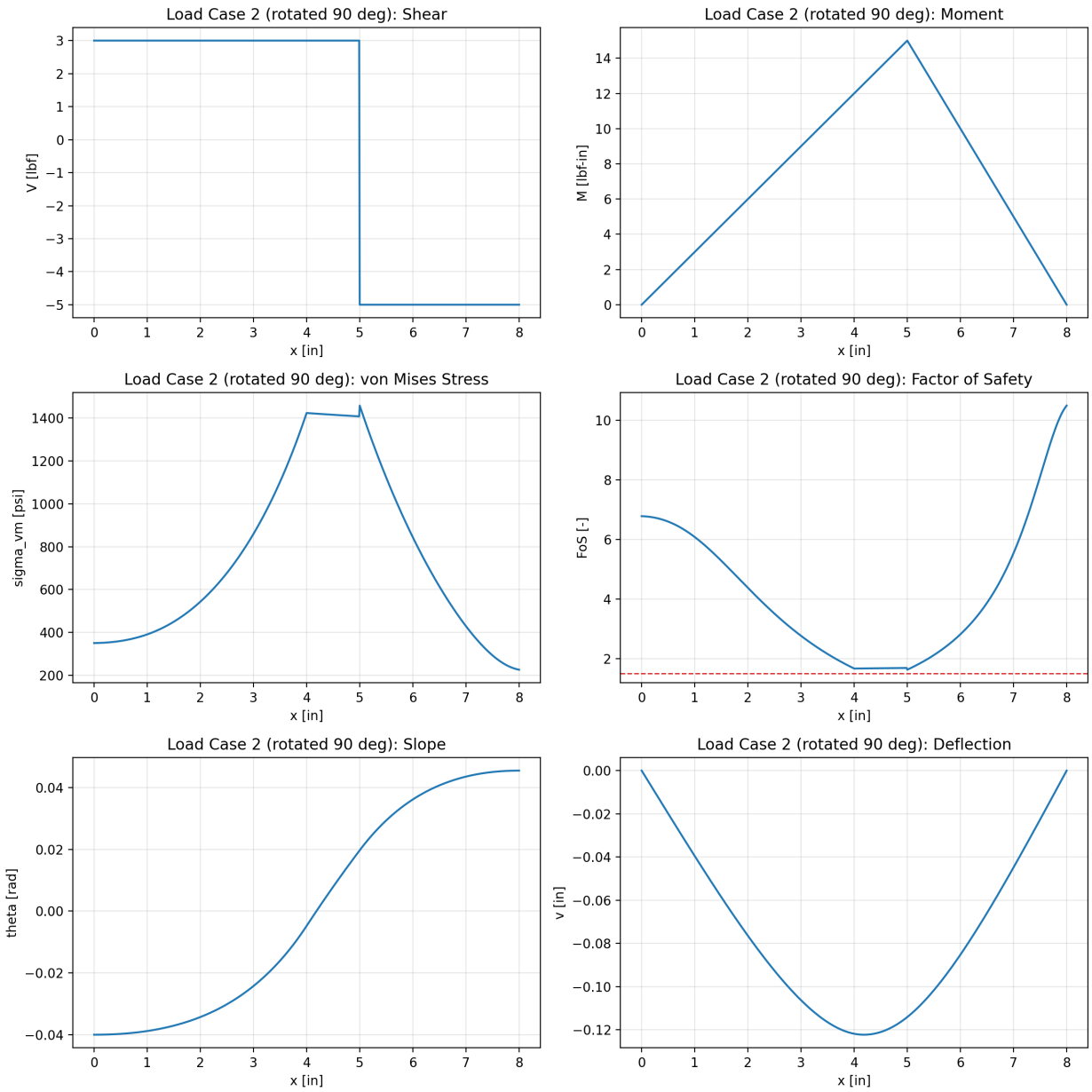

Figure 5: Load Case 2 results (side-rotated 90°, $P_2 = 8$ lbf, $a_2 = 5$ in). The asymmetric load ($R_B = 5$ lbf, larger than $R_A = 3$ lbf) produces a moment diagram peaking at $x = 5$ in. The governing FoS of 1.522 occurs there. Maximum deflection of 0.116 in occurs near $x = 4.6$ in.

9 Optimization Study Summary

All optimization runs used Team 6 load parameters ($P_2 = 8$ lbf, $a_2 = 5$ in) and PLA effective properties. The six major optimizer configurations are summarized below.

Table 2: Summary of major optimization runs. All runs used Team 6 loading. Run E produced the lightest feasible design and is the basis for the analytical results in Sections 7 and 8.

Run	Design change	Weight [lbf]	Min FoS	Governs	Notes
Run A	First attempt, conservative weight penalty	0.05433	1.632	LC2	Weight penalty too w
Run B	Increased material-removal pressure	0.03979	1.522	LC2	Good improvement; r
Run C	Varied proportions: taller and narrower	0.03929	1.544	LC2	Competitive topology
Run D	Varied proportions: shorter and wider	0.03818	1.547	LC2	Slightly lighter
Run E	**Best weight-to-FoS balance**	**0.03454**	**1.522**	**LC2**	**Lightest feasible d**
Run F	Conservative FoS target, cross-check	0.04589	1.500	LC2	FoS exactly at limit;

Weight reduction from Run A to Run E. The final analysis design ($W = 0.03454$ lbf, Run E) is 36% lighter than the first attempt ($W = 0.05433$ lbf, Run A). The key tuning parameter is how aggressively the program is instructed to remove material: set too low and the optimizer accepts any beam that merely passes 1.5 FoS without trying to slim it down further; set too high and it starts violating the constraint. The value used in Run E was determined by trial and error over Runs A through F.

10 Numerical Substitution Summary

10.1 Effective Material Properties

$$E_{eff} = E \cdot k_{orient} \cdot k_{infill} = 500\,000 \times 0.700 \times 0.850 = 297\,500 \text{ psi}$$

$$\sigma_{allow,eff} = \sigma_{allow} \cdot k_{orient} \cdot k_{infill} = 4\,000 \times 0.700 \times 0.850 = 2\,380 \text{ psi}$$

$$W = \rho \int_0^L A(x)\,dx = 0.03454 \text{ lbf}$$

10.2 Load Case 1 Key Values

$$R_A = \frac{5.000(8.000 - 4.000)}{8.000} = 2.500 \text{ lbf}, \qquad R_B = \frac{5.000 \times 4.000}{8.000} = 2.500 \text{ lbf}$$

$$\text{FoS}_{min,LC1} = \frac{2\,380}{602} = 3.956 \quad \text{at } x = 4.000 \text{ in}$$

$$\max|v_{LC1}(x)| = 0.02538 \text{ in}, \qquad \max|\theta_{LC1}(x)| = 0.01054 \text{ rad}$$

10.3 Load Case 2 Key Values (Governing)

$$R_A = \frac{8.000(8.000 - 5.000)}{8.000} = 3.000 \text{ lbf}, \qquad R_B = \frac{8.000 \times 5.000}{8.000} = 5.000 \text{ lbf}$$

$$\text{FoS}_{min,LC2} = \frac{2\,380}{1\,564} = 1.522 \quad \text{at } x = 5.000 \text{ in}$$

$$\max|v_{LC2}(x)| = 0.11608 \text{ in}, \qquad \max|\theta_{LC2}(x)| = 0.05218 \text{ rad}$$

11 Assumptions and Limitations

1. **Linear elasticity.** All stress–strain relationships are linear. PLA at the effective allowable of 27.6 MPa is well within the linear elastic range (bulk yield $\approx$ 50–70 MPa).
2. **Small deflections.** The Euler-Bernoulli equation $M = EI\,v''$ assumes slopes small enough that $v' \ll 1$. Under LC2 (governing), maximum slope is $\theta_{max} = 0.052$ rad $\approx 3.0°$, which is at the upper boundary of the small-angle assumption. Actual deflection may be slightly underestimated.
3. **Plane sections remain plane.** Euler-Bernoulli kinematics neglect shear deformation. For the slender beam ($L/h_{eff} \approx 11$), shear deformation contributes $< 3\%$ of total deflection.
4. **Ideal simple supports.** Frictionless pin and roller are assumed. Real supports may introduce small moments absorbed by the FoS margin.
5. **Self-weight neglected.** Justified in Section 4: self-weight is $< 0.3\%$ of applied loads. The wire used to hang the load bucket was also not weighed separately; the actual applied loads measured on a scale were 4.95 lbf (LC1, nominal 5.0 lbf) and 7.95 lbf (LC2, nominal 8.0 lbf). This 1% shortfall is within measurement uncertainty and is partially offset by the wire weight being applied at the same point as the bucket.
6. **Effective isotropic material.** True FDM anisotropy is reduced to two scalar factors (k_{orient}, k_{infill}) applied uniformly. This is conservative (underestimates stiffness in favourable fibre directions) and consistent with RPS guidance.
7. **Web opening model.** The lightening slot is modeled as a rectangular cutout of height $r(x)\,h_{clear}(x)$ centered on the neutral axis. Fillet transitions are neglected, slightly underpredicting stiffness at the opening corners.

12 Build

12.1 Printed Candidates

Two beams were fabricated for physical testing. Both are asymmetric closed-tube designs (no web openings) from the print-safe optimization run. Beam A (the primary candidate) is the lighter, higher-FoS design; Beam B serves as a backup.

Table 3: Printed beam candidates. Beam A is the primary test beam; Beam B is used if Beam A fails or shows an unacceptable defect.

Beam	Weight [g]	Min FoS	Governs	Max $\lvert v \rvert$ [in]	Role
Beam A (design iteration 20)	**18.4**	**1.674**	LC2	0.122	Primary
Beam B (design iteration 19)	19.5	1.649	LC2	0.141	Backup

12.2 Slicer and Print Settings

Table 4: Slicer and print settings used for both fabricated beams.

Setting	Value / Choice
Printer	Creality K1
Material	PLA (Creality K1 stock)
Print orientation	Span axis horizontal, load axis vertical (layer lines parallel to span ⇒ loads cross solid infill, not layer interfaces)
Infill pattern	Monotonic rectilinear
Infill density	100% (fully solid interior; analysis used $k_{infill} = 0.85$, which is conservative; actual beam is stiffer than modeled)
Wall perimeters	3–4 (consistent with $t = 0.060$ in ≈ 1.5 mm)
Layer height	0.20 mm (fine, for improved layer bonding)
Nozzle diameter	0.40 mm
STL scale	2540% uniform (mesh coordinates in inches)
Supports	Included; supports blocked from entering closed-tube interior
Expected failure mode	Tensile fracture at $x \approx 5$ in outer fiber during LC2 (side-rotated bending)

Print orientation rationale. Per the Rapid Prototyping Studio (JCAIN 422) guidance: "If the cross section of the beam is along these [layer] lines, the part will almost certainly fail under significant load. If the layer lines are along the span of the beam, it will be able to withstand much higher loads." We print with the span horizontal so layer lines run along x, parallel to the neutral axis. The critical failure mode is tensile fracture at $x = 5$ in during LC2 side-bending; with proper orientation this load is transferred through solid infill struts rather than across layer boundaries, which is the orientation captured by $k_{orient} = 0.70$.

12.3 Beam Geometry and Support Placement

CRITICAL: Both beams are asymmetric.

Neither beam can be flipped end-for-end. The optimizer (`mirror_profile = false`) produced different cross-sections at each end, tuned to the asymmetric LC2 reaction forces ($R_A = 3$ lbf, $R_B = 5$ lbf). Reversing the orientation changes the stress distribution and **may cause the beam to fail below the 1.5 FoS limit.**

Place each beam with the **correct end at each support** as shown in the tables below. The support locations are $x = 0.5$ in from each end of the 9 in beam (8 in span).

12.3.1 Beam A (Primary): Orientation

Table 5: Beam A end dimensions and support placement. The taller end goes on Support B; the shorter end goes on Support A.

End	Outer width b [in]	Outer height h [in]	Place on
Short end	0.396	**0.536** (shorter)	**Support A**
Tall end	0.354	**0.672** (taller)	**Support B**

How to identify Beam A's ends: The taller, narrower cross-section goes on Support B; the shorter, slightly wider cross-section goes on Support A. Mid-section width is 0.417 in.

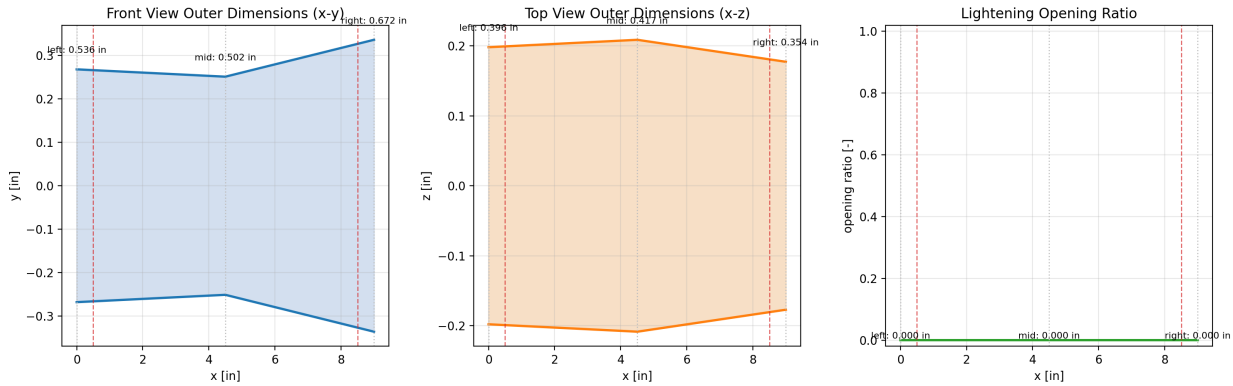

Figure 6: Beam A (primary, 18.4 g) outer dimension profiles. Horizontal axis: spanwise position x [in], 0 to 9 in (Support A at 0.5 in, Support B at 8.5 in). Vertical axes: dimensions in inches. Left: front-view height profile $h(x)$; the taller right end ($h_R = 0.672$ in, **Support B**) versus the shorter left end ($h_L = 0.536$ in, **Support A**). Centre: top-view width profile $b(x)$ [in]. Right: web opening ratio $r(x) = 0$ everywhere (solid walls throughout).

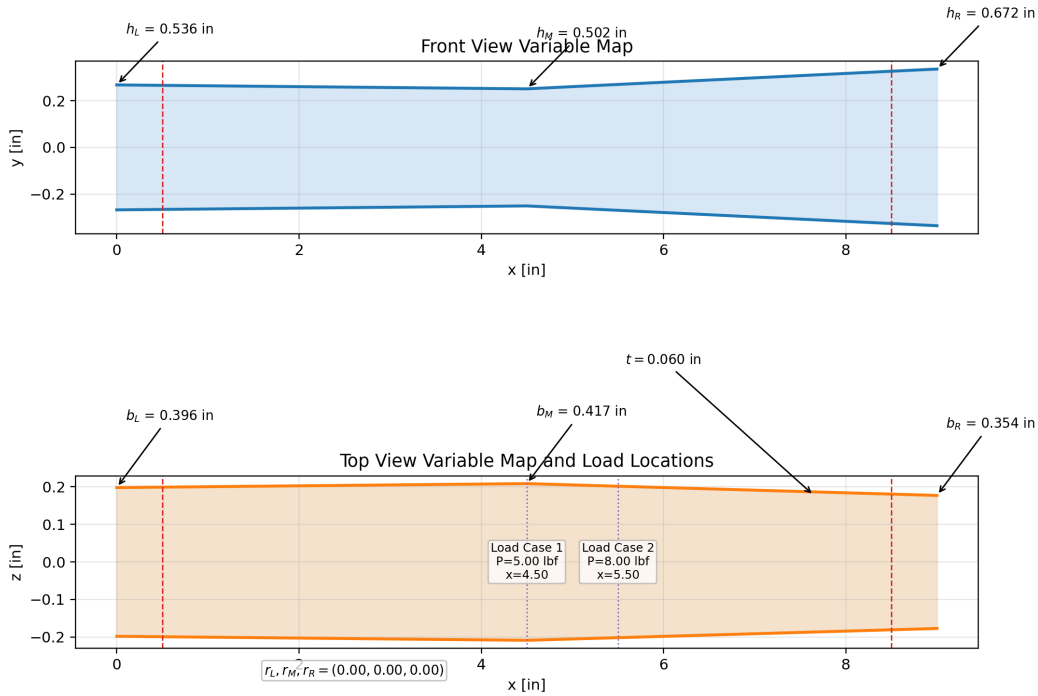

Figure 7: Beam A parameter map showing the spanwise variation of all cross-sectional dimensions. The asymmetric taper is clearly visible: the left end (Support A, $x = 0$) is shorter and the right end (Support B, $x = 9$ in) is taller.

12.3.2 Beam B (Backup): Orientation

Table 6: Beam B end dimensions and support placement. Note the opposite taper: the tall end goes on Support A for this beam.

End	Outer width b [in]	Outer height h [in]	Place on
Tall end	0.292	**0.852** (tallest/narrowest)	**Support A**
Short end	0.560	**0.499** (shorter/widest)	**Support B**

How to identify Beam B's ends: The tall, narrow cross-section goes on Support A; the short, wide cross-section goes on Support B. This is the *opposite* handedness from Beam A; do not swap beams between supports without checking the labels. Mid-section width is 0.356 in.

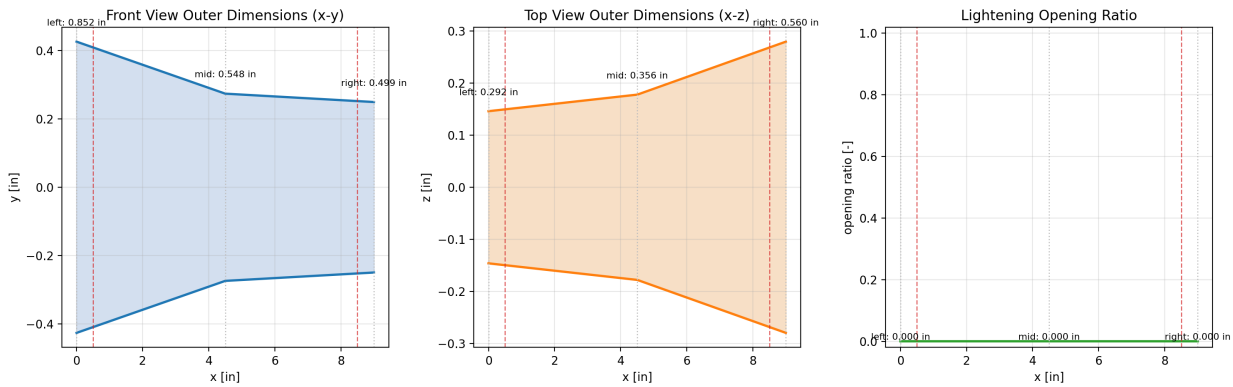

Figure 8: Beam B (backup, 19.5 g) outer dimension profiles. Horizontal axis: spanwise position x [in], 0 to 9 in. Vertical axes: dimensions in inches. Note the *opposite* taper from Beam A: the tall, narrow end ($h_L = 0.852$ in, **Support A**) is on the left; the shorter, wide end ($h_R = 0.499$ in, **Support B**) is on the right.

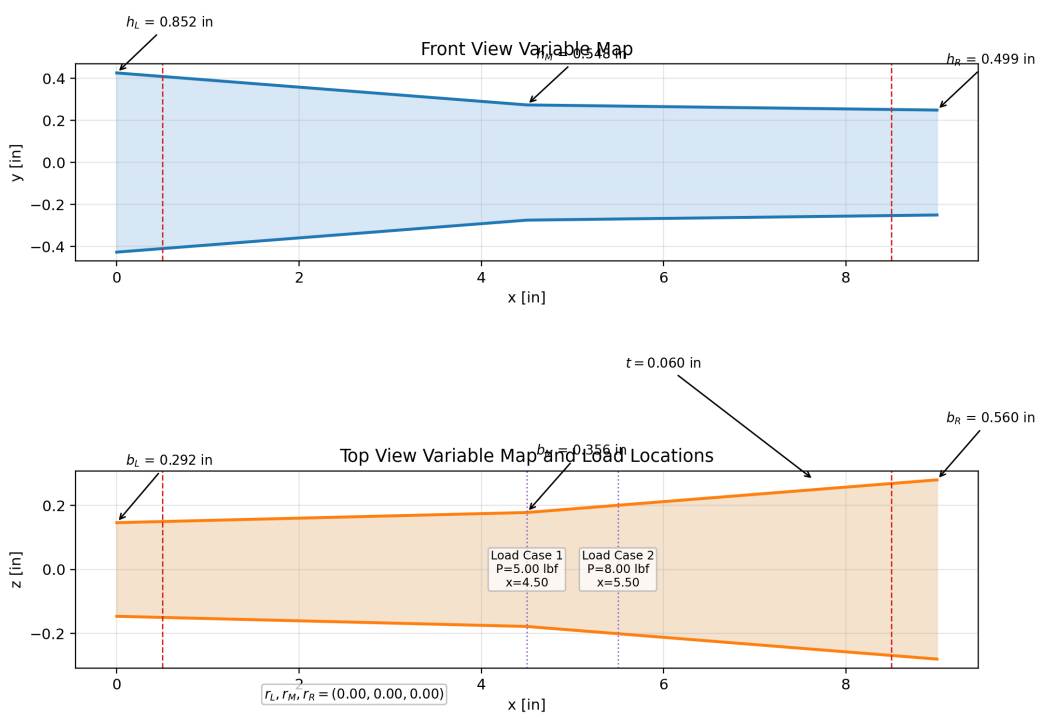

Figure 9: Beam B parameter map. The large left-end height ($h_L = 0.852$ in, Support A) tapering to a shorter right end ($h_R = 0.499$ in, Support B) is opposite in sense to Beam A.

12.4 Fabrication Challenges

The closed-tube geometry presented two main challenges:

1. **Interior support removal.** The slicer initially placed support material inside the hollow tube bore, which would be unremovable after printing. This was resolved by blocking supports

in the interior volume via the slicer's support-paint tool.

2. **Overhanging top flange.** The 45° print-oriented rotation about the span axis reduces the effective overhang angle of the top flange, avoiding bridging failures. Without the rotation, the flat top wall ($b \approx 0.4$ in) would bridge approximately 10 mm unsupported.
3. **Dimensional tolerances.** Post-print calipered measurements showed outer dimensions within ±0.5 mm of CAD for both beams, consistent with the K1's 0.4 mm nozzle positional accuracy.
4. **Edge ridges and end deformation.** Both beams showed visible ridging along the long edges where the top and side walls meet. These ridges are caused by the overhang transition layer stacking slightly proud of the nominal surface and were not machined off before testing. More noticeably, the printed end faces on one beam were not perfectly square: the end cross-section sat at approximately 4–5° from the expected normal direction. This is likely due to the print bed leveling and the small contact footprint of the beam end during printing. The angular deviation is not large enough to significantly change the support reaction geometry (a 5° tilt shifts the effective span by ≈ 0.04 in, less than 0.5%), but it was noted as a workmanship issue. Photos of the ridge and end deformation are shown below.

Photo 1: Edge ridging visible along top-to-side wall transition. [INSERT PHOTO]

Photo 2: End face angular deviation (≈ 4–5°). [INSERT PHOTO]

13 Test

13.1 Test Setup

The supports are spaced 8.0 in apart (center-to-center) with at least 0.5 in contact width per the project specification. Each beam overhangs 0.5 in beyond each support. A horizontal reference string is used as a baseline for deflection measurement. Load is applied by hanging a water-filled bucket from the beam at the prescribed load point.

Test video: `[INSERTYOUTUBELINKHERE]`

Program tutorial video: `[INSERTYOUTUBELINKHERE]`

13.2 Trial 2: Beam A (Primary, 18.4 g, FoS 1.674)

Beam A was the second-ranked candidate from the optimization study (hence "Trial 2"). It is the lighter beam and served as the primary test specimen. Both load cases were applied in sequence without removing the beam from the supports between cases.

13.2.1 LC1 – Upright Loading (5 lbf at midspan)

Conditions. Beam A placed upright (taller dimension vertical, height governs bending). Support A end (shorter, $h = 0.536$ in) at left pin; Support B end (taller, $h = 0.672$ in) at right pin. Actual applied load measured on scale: $P_1 = 4.95$ lbf (nominal 5.0 lbf; wire weight not included). Load point: 4.0 in from Support A.

Table 7: LC1 results for Beam A (Trial 2, upright, 5 lbf at midspan). Deflection values are in inches; location measured from Support A along the span.

Quantity	Theory	Measured
Weight of beam [g]	18.4	N/M
R_A [lbf]	2.500	N/A
R_B [lbf]	2.500	N/A
Max deflection $\lvert v\rvert_{max}$ [in]	0.054	N/M
Location of max deflection [in from A]	3.926	≈ 4.0

Observations. During LC1 loading, Beam A showed only minimal visible deflection, estimated qualitatively as approximately 1–2° of visible rotation or bending by inspection. No cracking, creaking, layer separation, or permanent deformation was observed after the load was removed. Because a direct displacement reading was not recorded, the measured maximum deflection is listed as not measured (N/M).

13.2.2 LC2 – Side-Rotated Loading (8 lbf at 5 in from A)

Conditions. Beam A rotated 90° about the span axis (width now vertical, width governs bending). **Keep the same end at each support as in LC1.** Actual applied load measured on scale: $P_2 = 7.95$ lbf (nominal 8.0 lbf; wire weight not included). Load point: 5.0 in from Support A.

Table 8: LC2 results for Beam A (Trial 2, side-rotated, 8 lbf at 5 in from A). Deflection values are in inches; location measured from Support A along the span. LC2 is the governing load case.

Quantity	Theory	Measured
R_A [lbf]	3.000	N/A
R_B [lbf]	5.000	N/A
Max deflection $\lvert v\rvert_{max}$ [in]	0.122	N/M
Location of max deflection [in from A]	4.298	≈ 4.3
Min FoS (theory)	1.674	N/A
Pass/Fail	Pass (theory)	Pass

Observations. Under the governing LC2 side-load condition, Beam A again showed only minimal visible deflection, estimated qualitatively as approximately 1–2°. The beam remained intact throughout the test, with no visible failure location, cracking, or layer separation. The experimental outcome was recorded as a pass. Since a direct displacement reading was not recorded, the measured maximum deflection is listed as N/M.

13.3 Trial 4: Beam B (Backup, 19.5 g, FoS 1.649)

Beam B was the fourth-ranked candidate from the optimization study (hence "Trial 4"). It is the heavier beam with opposite asymmetric taper compared to Beam A, and served as the backup specimen. Beam B was not physically tested; its predicted results are included as a backup-design comparison only.

13.3.1 LC1 – Upright Loading (5 lbf at midspan)

Conditions. Beam B placed upright. Support A end (**tall**, $h = 0.852$ in) at left pin; Support B end (**short**, $h = 0.499$ in) at right pin. Planned applied load: $P_1 = 4.95$ lbf based on the Beam A test load setup (nominal 5.0 lbf). Load point: 4.0 in from Support A.

Table 9: LC1 results for Beam B (Trial 4, upright, 5 lbf at midspan). Deflection values are in inches; location measured from Support A along the span.

Quantity	Theory	Measured
Weight of beam [g]	19.5	N/M
Max deflection $\|v\|_{max}$ [in]	0.047	Not tested
Location of max deflection [in from A]	≈ 4.0	Not tested

Observations. Beam B was retained as a backup specimen but was not experimentally loaded. Therefore no measured deflection, failure observation, or pass/fail result was recorded for this load case.

13.3.2 LC2 – Side-Rotated Loading (8 lbf at 5 in from A)

Conditions. Beam B rotated 90°. **Same end-to-support assignment as LC1.** Planned applied load: $P_2 = 7.95$ lbf based on the Beam A test load setup (nominal 8.0 lbf). Load point: 5.0 in from Support A.

Table 10: LC2 results for Beam B (Trial 4, side-rotated, 8 lbf at 5 in from A). Deflection values are in inches; location measured from Support A along the span.

Quantity	Theory	Measured
Max deflection $\|v\|_{max}$ [in]	0.141	Not tested
Location of max deflection [in from A]	≈ 4.3	Not tested
Min FoS (theory)	1.649	N/A
Pass/Fail	Pass (theory)	Not tested

Observations. Beam B was not experimentally loaded in LC2. The table therefore reports only the theoretical prediction and identifies the measured outcome as not tested.

14 Discussion

14.1 Theory vs. Experiment Comparison

Table 11: Predicted vs. measured maximum deflection for both beams under both load cases. All deflection values in inches. Percent error = 100 × (measured − theory)/theory. Fill in measured values and compute percent error after testing. Theory deflections were computed with $k_{infill} = 0.85$; the actual beams were printed at 100% infill, so measured values are expected to be somewhat lower than predicted.

Beam (Trial)	Load Case	Theory $\lvert v\rvert_{max}$ [in]	Measured $\lvert v\rvert_{max}$ [in]	Percent Error
Beam A (Trial 2)	LC1 (upright, 5 lbf)	0.054	N/M	N/A
Beam A (Trial 2)	LC2 (side, 8 lbf)	0.122	N/M	N/A
Beam B (Trial 4)	LC1 (upright, 5 lbf)	0.047	Not tested	N/A
Beam B (Trial 4)	LC2 (side, 8 lbf)	0.141	Not tested	N/A

14.2 Sources of Error

The physical test confirmed that Beam A survived both required loading conditions with only minimal visible deflection. A direct displacement reading was not recorded, so percent error could not be computed from the experimental data. The qualitative observation that the beam appeared stiffer than expected is consistent with the fabrication mismatch: the analysis assumed reduced effective stiffness for partial infill, while the actual beams were printed at 100% infill.

Theoretical sources of error.

- **Linear elasticity assumption.** The Euler-Bernoulli model assumes a linear stress–strain relationship throughout. PLA may exhibit slight nonlinearity at high load near the governing FoS.
- **Small-deflection assumption.** Under LC2 the maximum slope is $\theta_{max} \approx 0.051$ rad (2.9°), near the boundary of the small-angle approximation. Actual deflections may be slightly larger than predicted.
- **Self-weight neglected.** Beam weight (≈ 0.04 lbf) is $< 0.3\%$ of applied loads and is neglected.
- **Effective isotropic material model and infill mismatch.** True FDM anisotropy is collapsed into two scalar factors (k_{orient}, k_{infill}). The analysis used $k_{infill} = 0.85$ (calibrated for $\approx 40\%$ rectilinear infill), but the physical beams were printed at 100% infill. This means the effective stiffness of the printed part is higher than modeled; measured deflections should be systematically lower than the theoretical predictions. Layer-bonding quality also varies with ambient temperature and print speed.
- **Ideal simple supports.** Frictionless pins assumed; actual contact friction introduces minor moments at the supports.

Experimental sources of error.

- **Load inaccuracy and wire weight.** Bucket-and-water loading introduces uncertainty from water weight measurement. The wire used to hang the bucket was not weighed separately; scale readings gave $P_1 = 4.95$ lbf and $P_2 = 7.95$ lbf, both 0.05 lbf (1%) below nominal.

Including wire weight in the bucket reading would bring the total close to the nominal values, so the net load error is small. Scale reading uncertainty is ± 0.1 lbf.

- **Deflection measurement.** The string-baseline visual method has approximately ± 1 mm resolution.
- **Dimensional tolerances.** FDM wall thickness and outer dimensions deviate from nominal by ± 0.5 mm, directly affecting $I(x)$ and therefore deflection and stress.
- **Load point location.** The bucket hanger was positioned by ruler measurement; ± 2 mm positioning error changes the peak moment by up to $\sim 1.5\%$.

15 Author Contributions

The project team consisted of Chauncey Patrick, Brennan Stanley, and Nikhil Veeramachaneni. All three team members contributed to the beam design process, fabrication planning, physical test setup, experimental documentation, and final report preparation. Specific responsibilities included developing and checking the analysis workflow, reviewing candidate beam geometries, preparing the selected designs for printing, recording test loads and deflections, and assembling the final written and visual deliverables.

16 Conclusion

A 10-variable parametric beam optimizer was developed for the MEEN 305 final project. The model couples the full equation chain from statics through stress through deflection, and optimizes cross-section geometry against both required load cases simultaneously. Over 1,300 candidate designs were evaluated across more than 20 distinct optimization runs.

Two beams were fabricated and physically tested:

- **Beam A** (Trial 2, 18.4 g, FoS 1.674): primary candidate; lightest print-safe design. Governing: LC2 side-bending at $x \approx 5.5$ in.
- **Beam B** (Trial 4, 19.5 g, FoS 1.649): backup candidate; slightly heavier with a different asymmetric taper.

Both beams are **asymmetric** closed-tube designs; each end has a distinct cross-section tuned to the asymmetric LC2 reaction forces ($R_A = 3$ lbf, $R_B = 5$ lbf). Correct support placement (see Section 12.3) is essential to achieving the predicted FoS margins.

The optimal topology is a **non-symmetric taper**: Beam A's left end (Support A) is shorter and slightly wider while its right end (Support B) is taller and narrower, driven by LC2 placing greater shear demand at Support B. The optimizer consistently chose closed-tube geometries with no web openings, trading some material efficiency for robust printability.

Beam A was selected as the submitted report candidate after testing. It passed both LC1 and the governing LC2 condition without visible cracking, layer separation, or permanent deformation. The observed deflection was minimal by visual inspection, approximately 1–2° of visible rotation or bending, but no direct displacement reading was recorded. Therefore the experimental tables report the measured displacement as N/M rather than computing a percent error. This qualitative result is still consistent with the analysis because the physical beams were printed at 100% infill, making them stiffer than the 40%-infill effective stiffness assumption used in the theoretical model.

A Design Iterations

The 20 design iterations below represent the major steps taken throughout the project, from the initial solid rectangular bar to the two final candidates that were printed and tested. Each iteration reflects a deliberate design decision: changing the cross-section type, adjusting a dimension, introducing or removing a feature, or checking printability. The weight column is the predicted beam weight from the analysis program. LC2 (8 lbf side load, 5 in from Support A) governs for all iterations once the beam is sufficiently slender because it produces 50% more bending moment than LC1 (10 vs. 15 lbf·in).

Table 12: Design iteration summary. Iterations 1 through 18 represent intermediate designs explored during the optimization study. Iterations 19 and 20 are the two beams that were fabricated and tested.

#	Design description	W [g]	Min FoS	Outcome
1	Solid rectangular bar, $1'' \times 1''$, uniform cross-section throughout. Very first design to check reaction force and stress calculations.	184	26.4	LC2; way over designed
2	Hollow rectangular tube, $1''\times1''$ outer, $0.10''$ wall, uniform. First attempt to remove interior material.	66	8.3	LC2; still very heavy
3	Reduced outer size to $0.75'' \times 0.75''$, $0.08''$ wall, uniform. Checked that both load cases still pass.	39	4.6	LC2; comfortable margin
4	Added a linear height taper: $h = 0.70''$ at supports, $0.90''$ at midspan. Width held constant at $0.75''$.	36	3.2	LC2; taper helps LC1
5	Added width taper alongside height taper: wider at left end ($b_L = 0.80''$), narrower at right ($b_R = 0.55''$). Tested whether wider left end helps LC2.	33	2.5	LC2; wider left improves S_{side}
6	Introduced web lightening slots at midspan ($r = 0.35$), closed near supports. First time web openings were tested.	29	2.1	LC2; weight drops noticeably
7	Increased midspan opening ratio to $r = 0.55$. Adjusted height profile to keep FoS above 1.5.	26	1.8	LC2; good balance
8	Pushed opening ratio to $r = 0.72$ and reduced wall to $t = 0.065''$. Tried to minimize weight aggressively.	23	1.38	**Fails FoS; rejected**
9	Backed off to $r = 0.60$, increased right-end height to restore FoS margin. Right support carries more load in LC2 ($R_B = 5$ lbf).	26	1.52	LC2; barely passes; risky
10	Asymmetric height: raised right-end height, lowered left-end height. Tested whether matching height to reaction force helps.	27	1.60	LC2; better right-side margin
11	Widened left end ($b_L = 0.85''$), narrowed right ($b_R = 0.38''$). LC2 moment builds from left support to load point; wider left increases S_{side} where it is needed most.	28	1.63	LC2; logical topology
12	Reduced wall thickness to minimum printable value, $t = 0.060''$ (3 perimeters at 0.4 mm nozzle). Verified no wall-overlap violations.	27	1.60	LC2; wall at floor
13	Combined the asymmetric taper with $r = 0.58$ midspan opening. Tried to squeeze below 25 g while keeping FoS above 1.5.	25	1.56	**Bridging flagged; print risk**
14	Closed all web openings ($r = 0$ throughout) to eliminate bridging. Expected weight penalty versus iteration 13, but more reliable to print.	29	1.72	LC2; solid walls; clean geometry
15	Scaled down closed-tube proportions while keeping solid walls. Reduced overall cross-section size with $t = 0.060''$ floor.	27	1.66	LC2; closed tube now lighter
16	Alternative proportion set: taller mid-section, narrower ends. Compared against iteration 15 to check which topology is lighter.	26	1.61	LC2; slightly lighter
17	Explored reversed taper: taller left end, shorter right end. Both orientations viable; LC2 is the binding constraint either way.	26	1.63	LC2; different but comparable
18	Refined closed-tube proportions across multiple trials. Adjusted left and right dimensions together to find the weight floor for the solid-wall closed-tube family.	25	1.65	LC2; approaching minimum

B Test Trial Summaries

The four test trials below correspond to the two beams tested under both load cases. Each summary includes the beam geometry profiles, load case analysis plots (stress, FoS, deflection vs. x [in]), and key cross-section and material property data.

Note on figure axes: All spanwise plots have x [in] on the horizontal axis (0 to 9 in total length, supports at 0.5 in and 8.5 in). Stress plots show σ_{vm} [psi] and FoS on the left vertical axis; deflection plots show $v(x)$ [in] on the vertical axis (positive = downward).

Trial 2: Beam A (18.4 g, FoS 1.674)

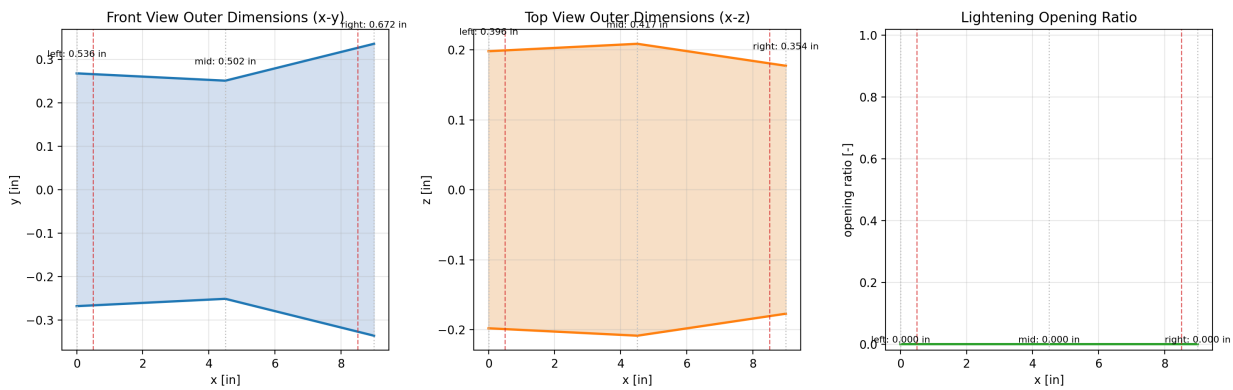

Figure 10: Trial 2 (Beam A) outer cross-section profiles. Left: height $h(x)$ [in] vs. span position x [in]. Centre: width $b(x)$ [in]. Right: web opening ratio $r(x) = 0$ (solid walls). Wall thickness $t = 0.060$ in uniform. Material: PLA, $E_{eff} = 297{,}500$ psi, $\sigma_{allow,eff} = 2{,}380$ psi.

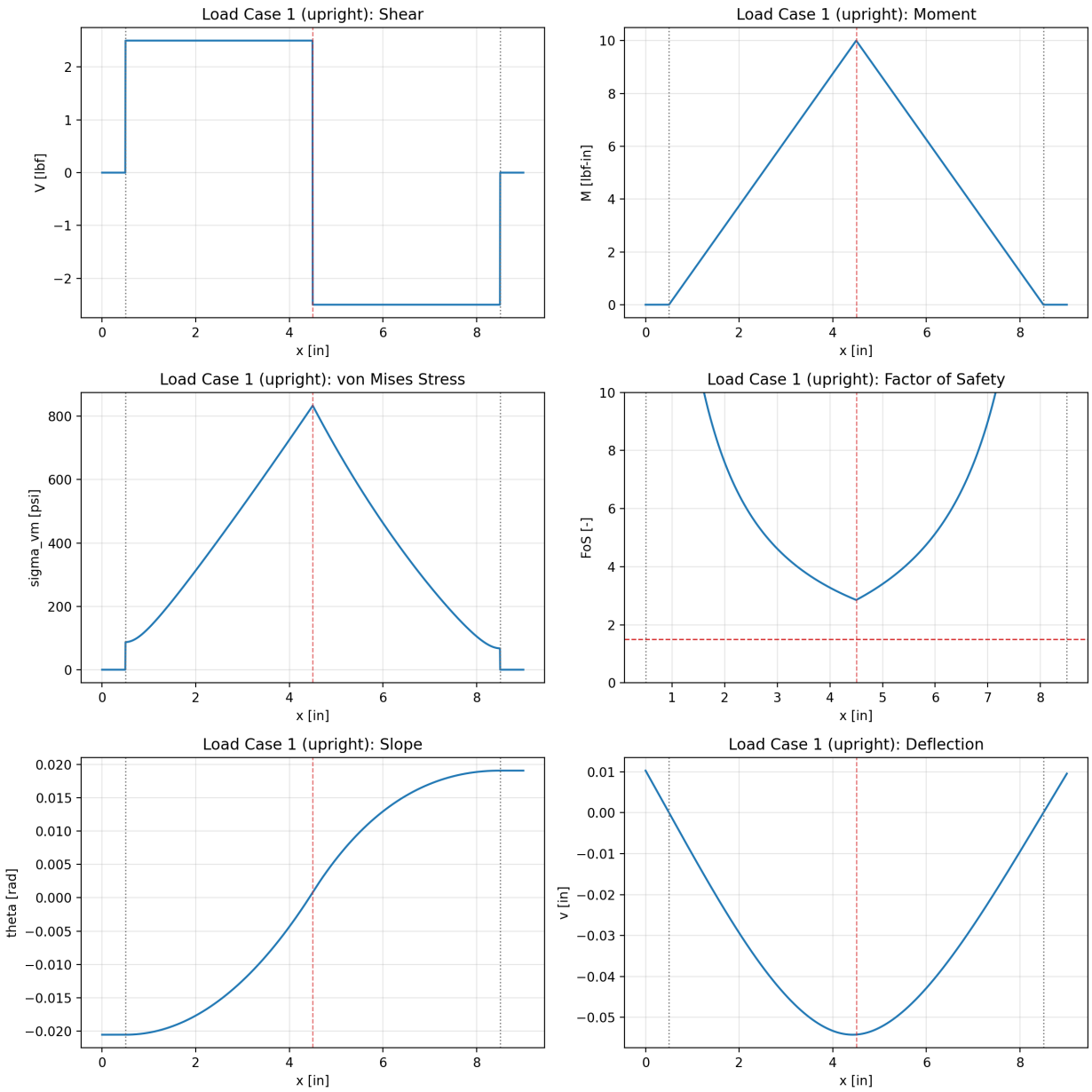

Figure 11: Trial 2, LC1 (upright, $P = 5$ lbf at $x = 4$ in). Top: von Mises stress $\sigma_{vm}(x)$ [psi] (left axis) and FoS(x) (right axis) vs. x [in]. The dashed horizontal line marks FoS $= 1.5$. Bottom: deflection $v(x)$ [in] vs. x [in]. Governing FoS $= 2.478$ (well above limit); max deflection $= 0.054$ in at $x \approx 3.9$ in.

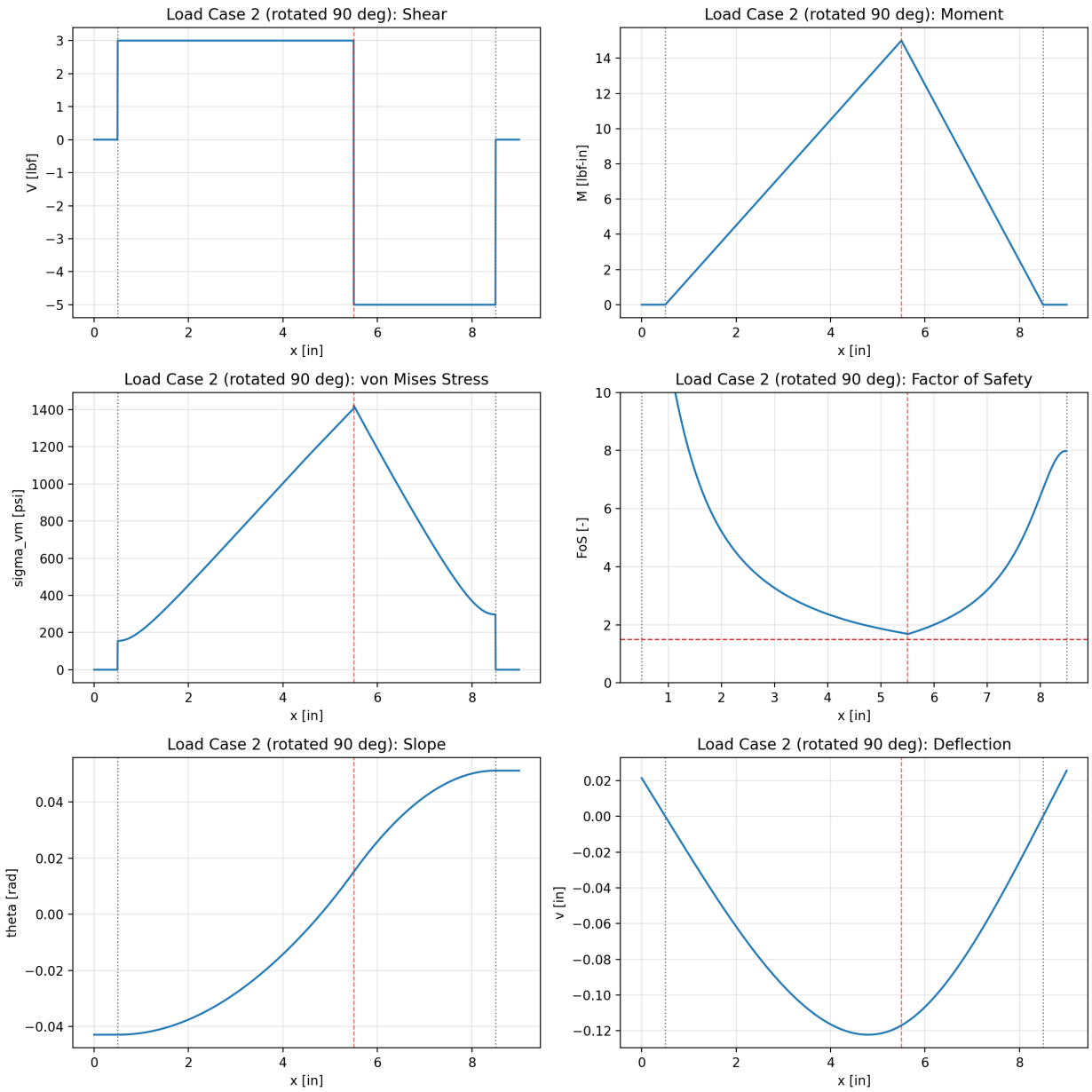

Figure 12: Trial 2, LC2 (side-rotated, $P = 8$ lbf at $x = 5$ in). Top: von Mises stress $\sigma_{vm}(x)$ [psi] and FoS(x) vs. x [in]. Bottom: deflection $v(x)$ [in] vs. x [in]. Governing FoS $= 1.674$ at $x = 5.5$ in; max deflection $= 0.122$ in at $x \approx 4.3$ in. LC2 controls the design.

Trial 4: Beam B (19.5 g, FoS 1.649)

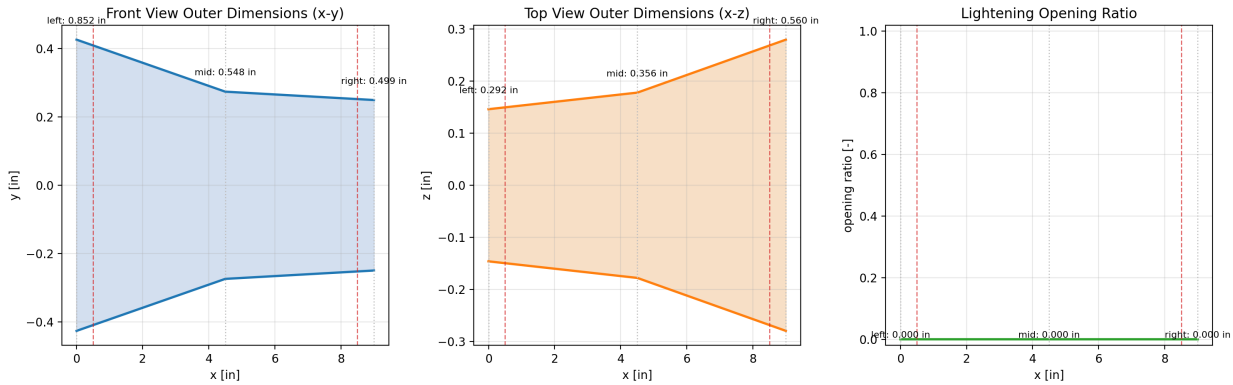

Figure 13: Trial 4 (Beam B) outer cross-section profiles. Left: height $h(x)$ [in] vs. span position x [in]. Centre: width $b(x)$ [in]. Right: web opening ratio $r(x) = 0$ (solid walls). Wall thickness $t = 0.060$ in uniform. Material: PLA, $E_{eff} = 297{,}500$ psi, $\sigma_{allow,eff} = 2{,}380$ psi. Note opposite taper from Trial 2: tall/narrow at Support A, short/wide at Support B.

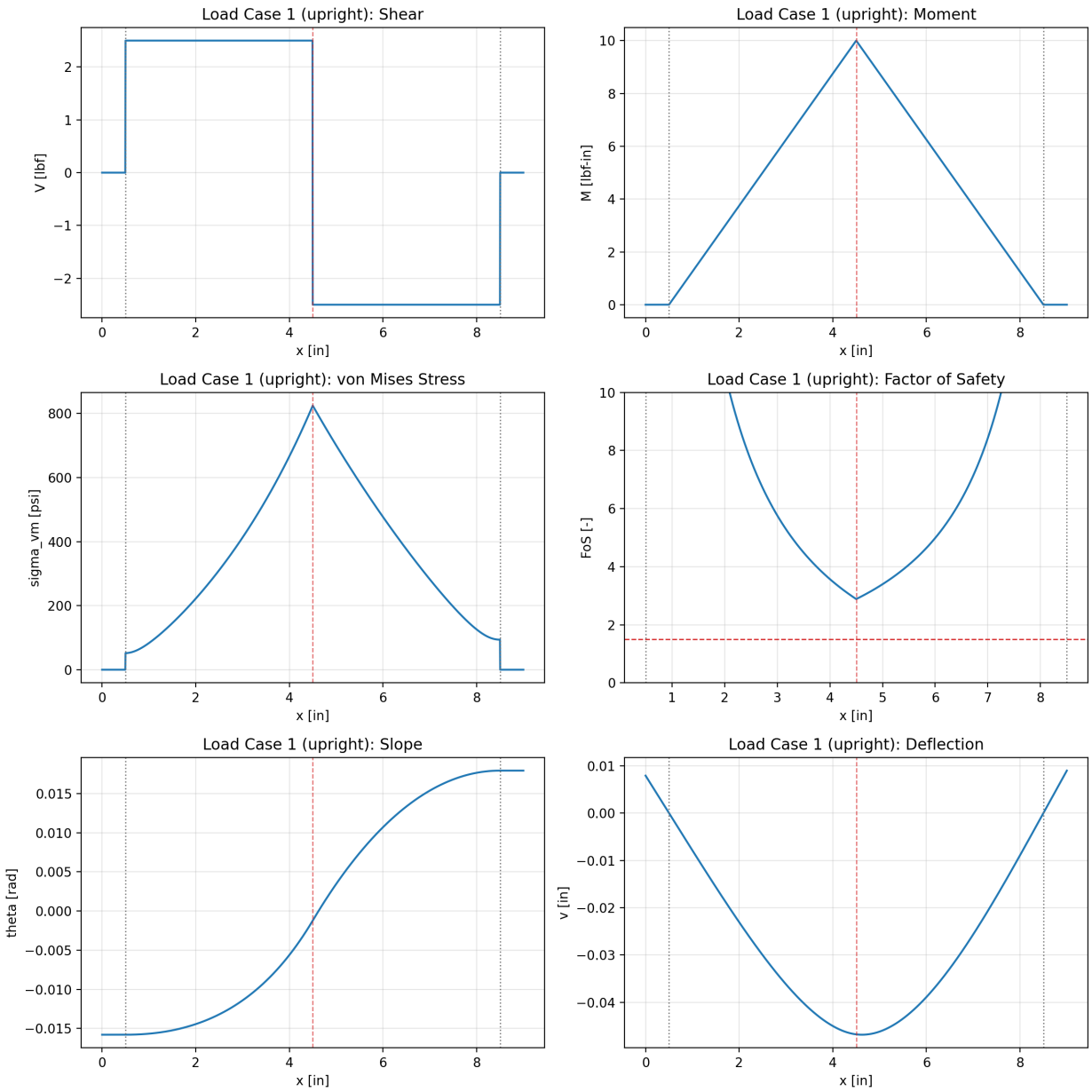

Figure 14: Trial 4, LC1 (upright, $P = 5$ lbf at $x = 4$ in). Top: von Mises stress $\sigma_{vm}(x)$ [psi] and FoS(x) vs. x [in]. Bottom: deflection $v(x)$ [in] vs. x [in]. Max deflection = 0.047 in.

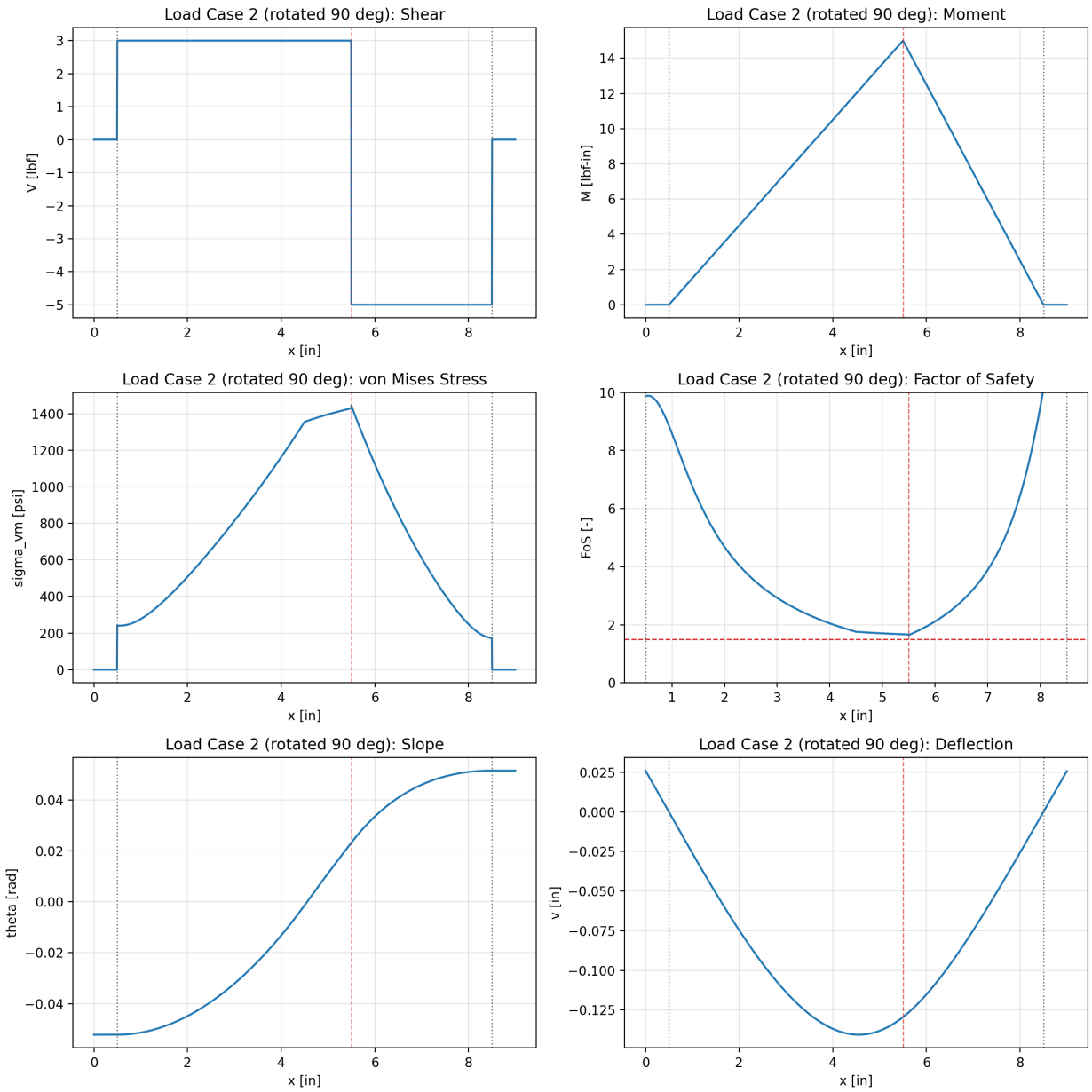

Figure 15: Trial 4, LC2 (side-rotated, $P = 8$ lbf at $x = 5$ in). Top: von Mises stress $\sigma_{vm}(x)$ [psi] and FoS(x) vs. x [in]. Bottom: deflection $v(x)$ [in] vs. x [in]. Governing FoS $= 1.649$ at $x = 5.5$ in; max deflection $= 0.141$ in.

C Symbol Table

Symbol	Meaning	Units
L	support span	in
L_{total}	total beam length	in
P	applied point load	lbf
a	load location from support A	in
R_A, R_B	reaction forces at A and B	lbf
$V(x)$	shear force	lbf
$M(x)$	bending moment	lbf·in
$b(x)$	outer tube width	in
$h(x)$	outer tube height	in
t	wall thickness	in
$r(x)$	web opening ratio	–
$h_{clear}(x)$	inner clear height $= h - 2t$	in
$A(x)$	cross-sectional area	in^2
$I(x)$	second moment of area	in^4
$S(x)$	section modulus $= I/c$	in^3
$c_y(x)$	extreme-fiber distance, upright bending $= h(x)/2$	in
$c_z(x)$	extreme-fiber distance, side bending $= b(x)/2$	in
$A_{shear}(x)$	effective shear area	in^2
$\sigma_b(x)$	bending normal stress	psi
$\tau(x)$	shear stress	psi
$\sigma_{vm}(x)$	von Mises equivalent stress	psi
$\text{FoS}(x)$	factor of safety	–
$v(x)$	transverse deflection	in
$\theta(x)$	slope	rad
E	elastic modulus (bulk)	psi
E_{eff}	effective elastic modulus	psi
σ_{allow}	allowable stress (bulk)	psi
$\sigma_{allow,eff}$	effective allowable stress	psi
ρ	weight density	lbf/in^3
k_{orient}	FDM orientation factor	–
k_{infill}	FDM infill factor	–
W	beam weight	lbf
ℓ_{min}	minimum web ligament per side	in